

OpenVZ is said to be rarely more than 3 per cent. So, on a server with a few gigs of RAM, it is possible to run tens of VPSs and still have decent performance. Since there is only one kernel to deal with, memory consumption is also under check.

User bean counters

OpenVZ is not just about the isolation of processes. There are various resources on a computer system that processes compete for. These are resources like CPU, memory, disk space and at a finer level, file descriptors, sockets, locked memory pages and disk blocks, among others. At a VPS level, it is possible in OpenVZ to let the administrator set limits for each of these items so that resources can be guaranteed to VPSs and also to ensure that no VPS can misuse available resources. OpenVZ developers have chosen about 20 parameters that can be tuned for each of the VPSs.

The OpenVZ fair scheduler

Just as various resources are guaranteed to VPSs, CPU time for a VPS can also be guaranteed. It is possible to specify the minimum CPU units a VPS will receive. To make sure this happens, OpenVZ employs a two-level scheduler. The first level fair scheduler makes sure that no VPS is starved of its minimum CPU guarantee time. It basically selects which VPS has to run on the CPU next. At the scheduler level, a VPS is just a set of processes. Then, this set is passed on to the regular Linux kernel scheduler and one from the set is scheduled to run. In a VPS Web hosting environment, the hosting provider can thus guarantee the customer some minimum CPU power.

Installing OpenVZ

To install OpenVZ and have it work, you need to download or build an OpenVZ kernel, and also build or download pre-built OpenVZ tools. When you install the OpenVZ tools, it also installs the *init* scripts that take care of setting up OpenVZ. During system start-up and shut down, VEs are automatically started and shut down along with the Hardware Node (HN). Once the tools are installed, you can see that a directory named 'vz' is created in the root directory and it also contains other directories. On a production server, you may want '/vz' on a separate partition.

Installing the kernel

There are a few options here. If you are running a CentOS or RHEL distribution of Linux, pre-built kernels are available. Although the current version of Linux is 2.6.29, OpenVZ stable kernels are still in the 2.6.18 series. These are considered stable and they are either based on RHEL/CentOS kernels or the vanilla *kernel.org* 2.5.18 kernel. If you are running CentOS or RHEL, you can expect these 'stable' kernels to play well with

your system. For the sake of demonstration, I will be explaining a source-based installation.

Since we are installing from source code, first you'll need to fetch the kernel from *kernel.org*. Later on, fetch the OpenVZ patch and the OpenVZ kernel config file required to configure the kernel. Get the 'i686-SMP' config file, if you are on a 32-bit machine.

```
$ wget http://www.kernel.org/pub/linux/kernel/v2.6/linux-2.6.18.tar.bz2
$ wget http://download.openvz.org/kernel/branches/2.6.18/028stab096.1/patches/patch-ovz028stab096.1-combined.gz
$ wget http://download.openvz.org/kernel/branches/2.6.18/028stab096.1/configs/kernel-2.6.18-i686-smp.config.ovz
```

Now, let's untar the kernel and apply the patch:

```
$ tar xvfj linux-2.6.18.tar.bz2
$ gunzip patch-ovz028stab096.1-combined.gz
$ cd linux-2.6.18
$ patch -p1 < ../patch-ovz028stab096.1-combined
```

Let us configure the kernel based on the OpenVZ config file and compile it:

```
$ cp ../kernel-2.6.18-i686-smp.config.ovz .config
$ make bzImage && make modules
```

Time now for installation:

```
$ sudo make modules_install
$ cp arch/i386/bzImage/boot/vmlinuz-2.6.18.openvz
$ cp .config/boot/config-2.6.18.openvz
$ cp System.map /boot/System.map-2.6.18.openvz
```

Create the initrd:

```
$ cd /boot
$ sudo mkinitramfs -o initrd.img-2.6.18.openvz 2.6.18-028stab096
```

Add the new kernel to the GRUB menu in */boot/grub/menu.lst*, and just append these lines:

```
title      OpenVZ
root       (hd0,0)
kernel    /boot/vmlinuz-2.6.18.openvz root=/dev/sda1 ro
initrd     /boot/initrd.img-2.6.18.openvz
savedefault
boot
```

Installing the OpenVZ tools

Now that the kernel installation is done, we can install the tools. The "vzctl" package contains the main utilities to manage OpenVZ virtual private servers, and the "vzquota" package contains utilities to manage disk quota. Installing these utilities is a simple affair, as